

Numerical Simulation of Accretion onto Compact Object with Hard Surface



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1. Introduction

- Compact objects like Neutron Stars and Black holes have a very prominent accretion disk.
- The flows that are present in this accretion disk is accelerated so much that as it goes near the star it achieves supersonic speeds.
- Unlike Black holes that have an event horizon, Neutron stars have a hard surface beyond which the flow cannot go.
- A flow which is supersonic would slam into the hard surface and creates a huge shockwave.
- Thus, the flow must slow down before reaching the surface.
- The braking of the flow can occur through shocks that allow it to jump from a supersonic flow to a subsonic flow.
- For these shocks to be stable, a boundary layer is required on the surface. ¹
- This boundary layer slows down the incoming matter and dissipates the energy from subsonic flows all the way down to zero before it reaches the surface.

2. Assumptions

- Weakly magnetized neutron star.
 - Non-viscous, non-dissipative flow.
 - Axisymmetric flow with constant angular momentum.
 - Paczynski-Wiita gravitational potential. ²
- $$\Phi_{PW} = -\frac{GM}{r - r_s}$$
- Relativistically correct Chattopadhyay&Ryu EoS. ³
(Allows the polytropic index to vary with temperature for multispecies flows)

3. Setup and Numerical Model

- Using the assumptions mentioned previously, an analytical accretion solution is obtained.
- 1-D hydrodynamical simulation using Roe-type Riemann solver is implemented with uniform grid.
- The units of length, speed and mass are Schwarzschild radius, speed of light and the mass of the blackhole.
- Inner boundary condition is the boundary layer model, with parameterized cooling function.
- Simulation is run till steady state is achieved.

5. Conclusions

- Multiple shocks can form in the flow that allows it to transition from supersonic to subsonic near the surface.
- There is a boundary layer on the Neutron star surface that dissipates some of the energy and cools the flow.
- The more energy that is dissipated, i.e. the cooler the flow near the surface is, the closer the shock can form.
- We can obtain the different shock solutions when the cooling parameter of the boundary layer is changed.

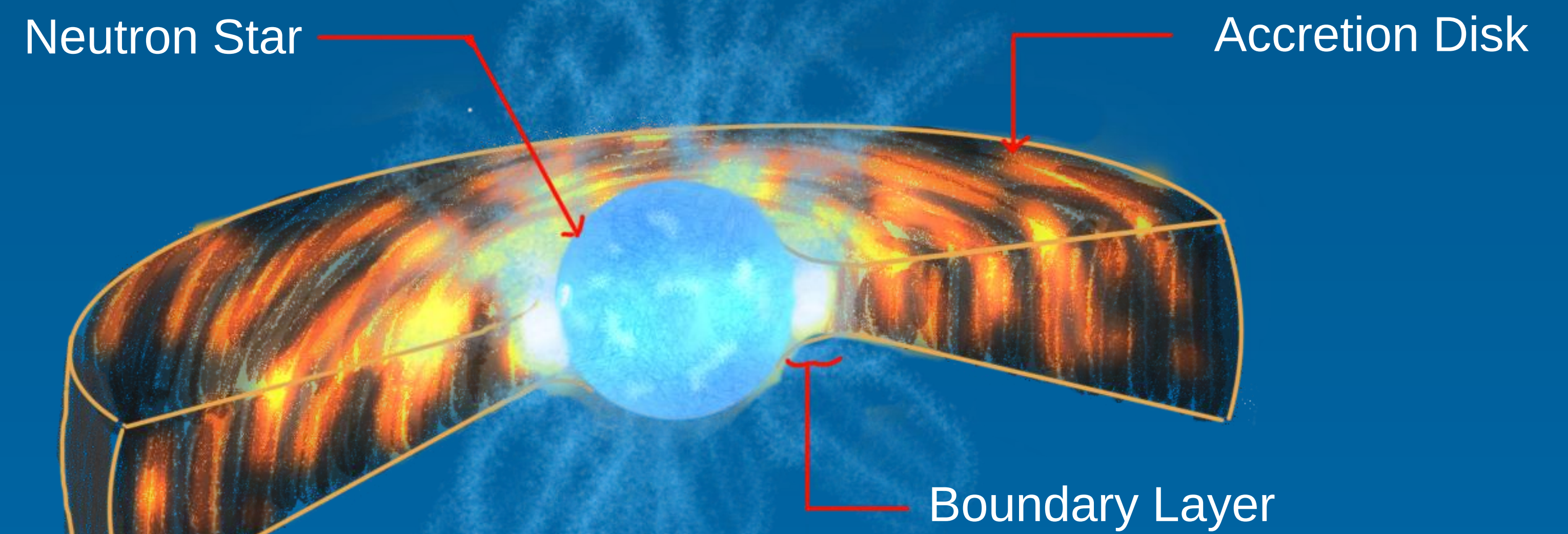


Fig.1: Weakly magnetized Neutron star with accretion disk cut away to show boundary layer

4. Results

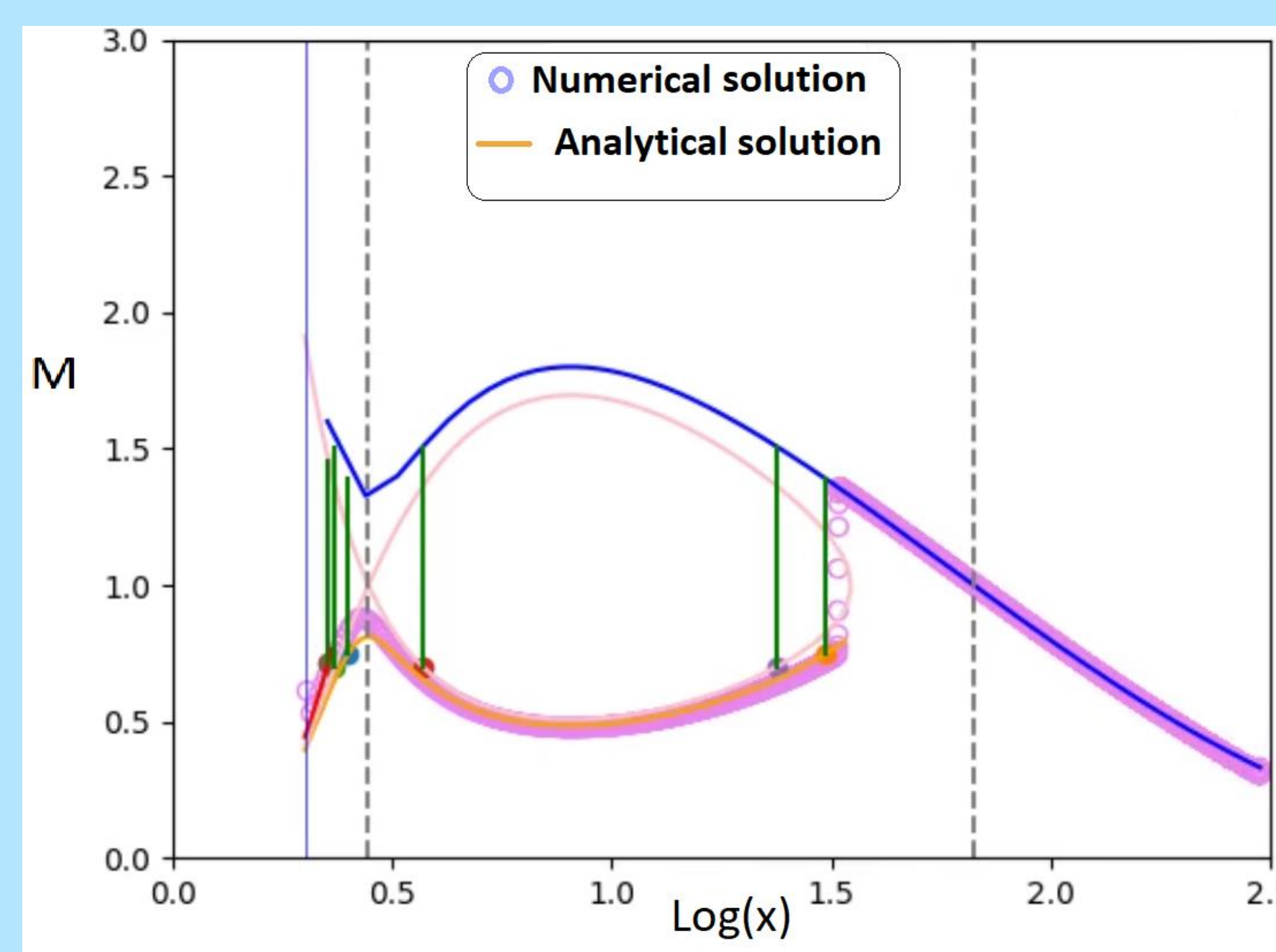


Fig.2a: High λ , Low E (Hotter boundary layer)

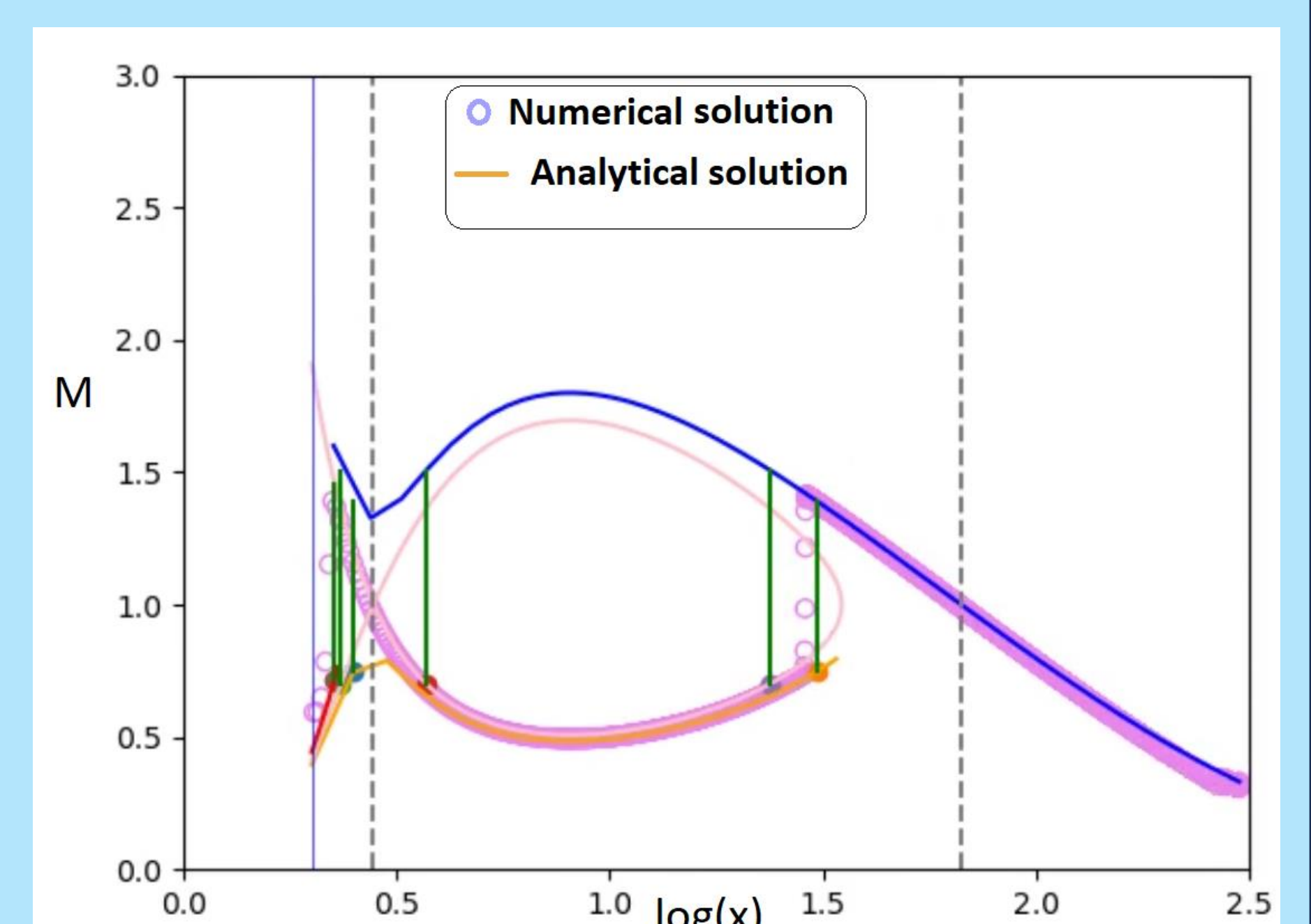


Fig.2b: High λ , Low E (Cooler boundary layer)

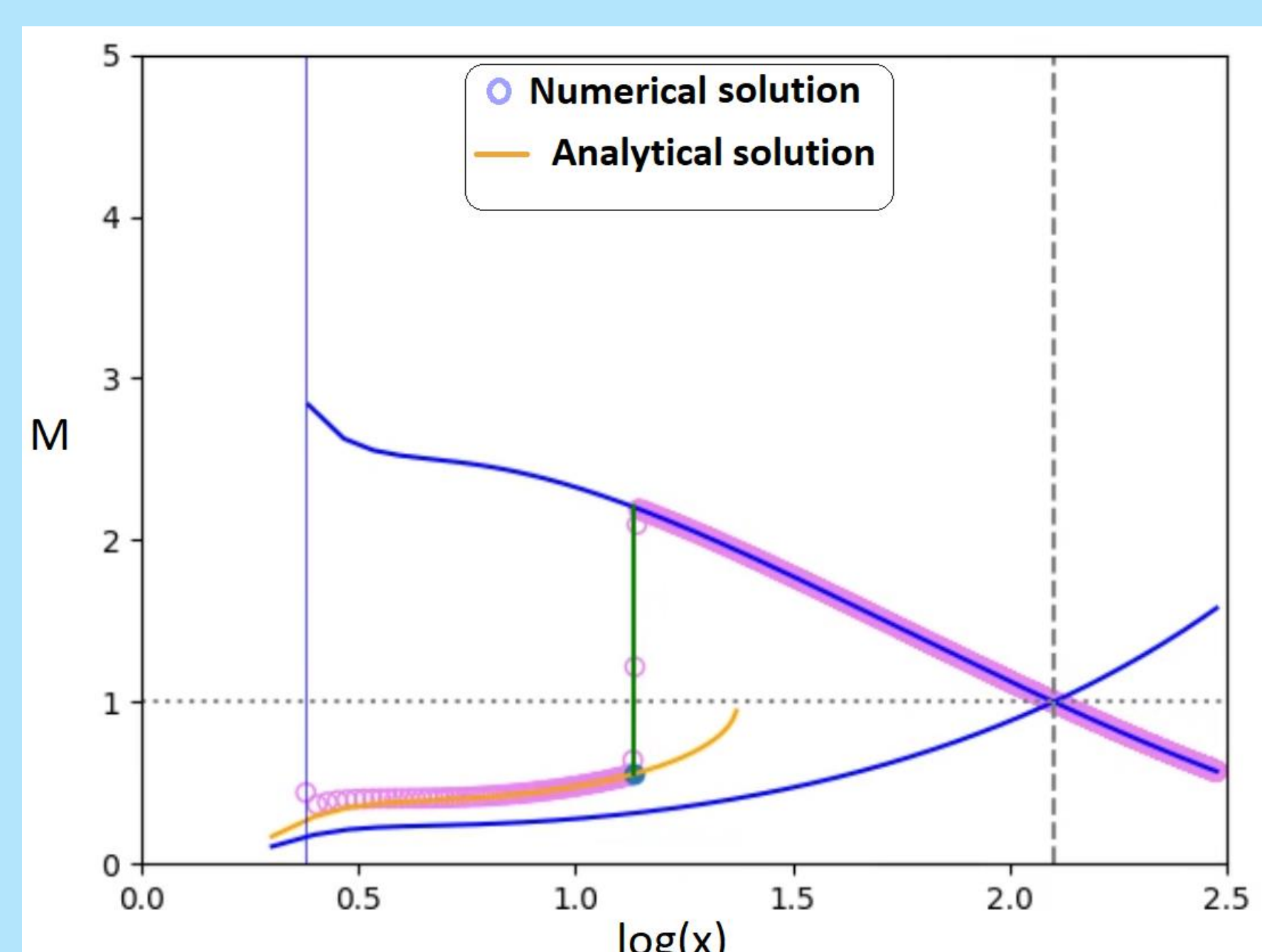


Fig.2c: Low λ , Low E

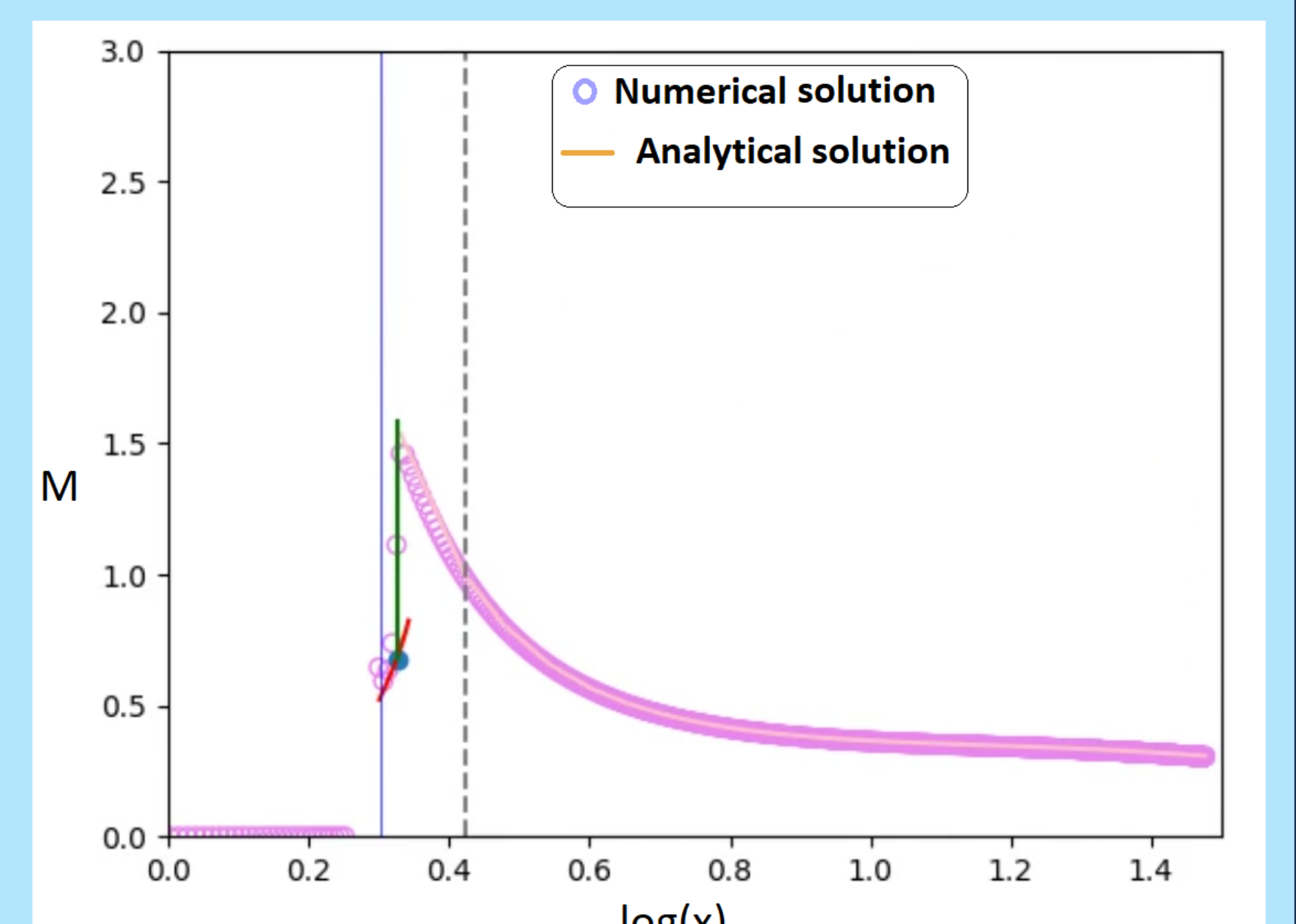


Fig.2d: High λ , High E

Fig.2: Subplots a,b,c and d show Mach number (M) vs logarithmic radial distance (x) for different types of solutions with varying angular momentum (λ) and specific energy (E). The purple overlay shows numerical solution on colored solid lines corresponding to analytical solutions where the vertical green lines depict possible shock-jumps. In Fig 2a and 2b, the same case with different hardness levels of the boundary layer is depicted.

6. Future works

- Improve Boundary Layer model to better correspond to physical models.
- Include radiation model to analyze the emission signatures of the flow, particularly in the X-rays.

7. References

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